

Written Status Report

CMSI 4072 Project Status Sheet Spring 2026

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Project: Sammy's Games: A Board and Card Game Platform

Date: March 11, 2026

Period Covered: February 27 - March 11

Accomplishments since last report:

Began and substantially completed the implementation of Six Hands, a solitaire card game played on a daily-generated grid of 30 cards. Specific accomplishments include:

- Designed and implemented a 6×5 card grid using HTML, CSS, and JavaScript, with daily card generation driven by a seeded pseudorandom number generator (mulberry32) so that all players receive the same cards each day.
- Implemented a color-based grouping system allowing players to assign cards to one of five colored groups using an interactive color picker, with live visual feedback on selection.
- Implemented a sixth-hand mechanic using a black color mode, allowing players to designate one card from each of the five groups to form a sixth five-card hand.
- Built a validation system that confirms all 30 cards are colored, each color group contains exactly 6 cards, and exactly one card per group is marked for the sixth hand, displaying a live VALID/INVALID status.
- Displayed all six resulting hands in a live side panel that updates dynamically as the player makes selections.
- Integrated Six Hands into the site navigation alongside the existing games.

Scheduled tasks to be done by next report:

- Implement orthogonal connectedness checking in the Six Hands validator, as the current VALID check does not yet enforce that each color group must be fully connected (March 25).
- Add poker hand strength evaluation so that each group's hand rank is displayed and players can identify their weakest hand at a glance (March 25).
- Polish the Six Hands UI for clarity and usability based on informal feedback (March 25).
- Add hand strength comparison to allow players to see how their arrangement stacks up (March 25).
- Continue improving Knockout bot decision-making beyond the defensive logic added this period (March 25).
- Begin scoping and planning a third game for the platform (March 25).

Noteworthy risks, concerns, or problems:

Risk: Timeline pressure remains a concern. Six Hands is functional but not yet fully complete, and the remaining features — connectedness validation, hand evaluation, and UI polish — represent meaningful additional work alongside continued Knockout improvements.

Mitigation Plan: Six Hands requires no AI opponent, keeping its remaining complexity manageable. Knockout bot improvements will be scoped conservatively to avoid overcommitting.

Concern: The connectedness rule is a core mechanic of Six Hands but is currently unenforced in the validator. Players can submit technically invalid groupings without being warned.

Mitigation Plan: Implement a flood-fill or BFS traversal on each color group to enforce orthogonal connectivity before a layout is marked VALID.